

UDITA RAHMAN ARSHEE

Electronics & Communication Engineering Graduate | Wireless Networks | WSN/IoT | AI for Communications

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RESEARCH INTERESTS

Wireless Sensor Networks • Internet of Things • Wireless Communications • Intelligent Wireless Systems • Network Optimization & Resource Allocation • Energy-Efficient Communication Systems • Distributed/Federated Learning • Intelligent Sensing

EDUCATION

B.Sc. Engineering in Electronics and Communication Engineering

[2022 - 2026]

Khulna University of Engineering and Technology, Khulna, Bangladesh

CGPA: 3.26/4.00

Final four term GPAs: 3.65, 3.74, 3.65, 3.70

Selected Coursework: Wireless Communications, Computer Networks, Digital Communication, Information Theory, Digital Signal Processing, Signal & Systems, Antenna Engineering, Microwave Engineering, Machine Learning, Microprocessors & Microcomputers, Telecommunication Engineering, Satellite Communications & RADAR.

RESEARCH EXPERIENCE

Spatially-Constrained Event-Driven Hierarchical Deep Reinforcement Learning (SCED-HRL) for Energy Optimization in Wireless Sensor Networks

Undergraduate Thesis, Department of ECE, KUET | Supervisor: Dr. Monir Hossen | 2026

- Developed a hierarchical DRL framework combining an event-driven PPO strategic layer with DDQN-based tactical agents for energy-aware WSN control.
- Introduced spatial energy budgeting and feasibility constraints to extend network lifetime and balance energy consumption across sensor regions.
- Evaluated on Intel Berkeley sensor data and synthetic traffic against baseline WSN control strategies on network lifetime, energy fairness, latency, and packet delivery; a first-author thesis-derived manuscript has been submitted to ICEEICT 2027.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Publications

[1] A. R. Aditta, P. K. Das, **U. R. Arshee** and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road-Safety and Biomedical Reliability Monitoring," *IEEE BECITHCON*, 2026.

Submitted Manuscripts

[2] **U. R. Arshee**, M. Hossen and S. I. Mumu, "Event-Driven Hierarchical Deep Reinforcement Learning with Spatial Energy Constraints for Wireless Sensor Network Optimization," submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2027)*. **[First Author]**

[3] S. I. Mumu, M. Hossen and **U. R. Arshee**, "THRIFT: Selective Graph-Teacher Knowledge Transfer for Federated Intrusion Detection in Wireless Sensor Networks," submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2027)*.

[4] Most. U. Jannat, **U. R. Arshee**, M. S. Tanvir, D. Aich, A. K. M. B. Haque, X. Xu and G. Muhammad, "Adaptive PSO-Weighted Grey Wolf Optimized Deep-Q Network Framework for Balancing Energy Consumption and Anomaly Detection Performance in Industrial Wireless Sensor Networks," submitted to the *IEEE Internet of Things Journal*, 2026.

[5] A. K. M. B. Haque, B. Bhushan, Most. U. Jannat, **U. R. Arshee** and P. Bhattacharya, "Deployment of Smart City Concept: Components, Architecture, Applications, Security Issues and Future Directions," manuscript submitted, 2026.

Manuscript Under Revision

[6] A. R. Aditta, **U. R. Arshee**, S. M. R. Islam and N. Taaha, "Transformer-Based Modeling of DNA Enhancers: A Controlled Evaluation of DNABERT and Longformer for Identification and Strength Prediction," under revision for journal resubmission, 2026.

SELECTED ACADEMIC PROJECTS

- **RFID-Based Smart Car Parking System:** IoT parking prototype using **RFID, ESP8266, Arduino** and **Blynk** for vehicle identification and monitoring.
- **Satellite Link Performance Analysis:** Evaluated link performance under varying frequencies, signal parameters and propagation distances, focusing on SNR and path loss.
- **Dual-Band Microstrip Patch Antenna:** Designed and simulated a patch antenna for dual-band operation at **3 GHz** and **7 GHz**.
- **Pre-emphasis & De-emphasis Filters:** Designed and analyzed filter characteristics in **MATLAB/Simulink** for telecommunication signal processing.
- **Wireless Communication Laboratory:** Hands-on experiments with GSM, Zigbee, Bluetooth and RFID platforms.

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost; Deep Reinforcement Learning (PPO, DDQN), CNNs, GNNs, Tree-Based Models, Model Evaluation
- **Programming & Scientific Computing:** Python, C, SQL, MATLAB, NumPy, Pandas, Matplotlib
- **Computer Networking:** Routing & Switching, VLAN Configuration, OSPF, ACLs, DHCP, DNS, Network Design & Troubleshooting, Cisco Packet Tracer
- **Embedded & IoT Systems:** Raspberry Pi, ESP32, ESP8266, Arduino, RFID, GSM, Zigbee, Bluetooth, OpenCV
- **Engineering Tools:** Simulink, Multisim, Arduino IDE, Visual Studio Code

HONORS & AWARDS

- **Champion**, Technical Writing, EUREKA S/1, **KUET Research Society** [2025]

ACADEMIC & RESEARCH LEADERSHIP

Executive, KUET Research Society [2025-2026]

- Organized a department-wide pre-mid thesis competition involving 3rd- and 4th-year ECE students to promote research presentation, peer learning and early thesis engagement.
- Coordinated online graduate-study and research guidance sessions with KUET alumni pursuing M.S./Ph.D. degrees abroad, helping current students understand research preparation, graduate applications, and academic career pathways.

Campus Ambassador, SheSTEM [2024-2026]

- Promoted STEM awareness and encouraged student engagement in technical initiatives and events.

CERTIFICATIONS & TRAINING

- **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI, Coursera
- **Computer Networking, EDGE Project**, ICT Division, Government of Bangladesh
- **Data Analysis & Programming**, DataCampData Analysis & Programming, DataCamp
- **Industrial Visits:** UGC National Data Center, Bangladesh Cable Shilpa Limited (BCSL), Walton Hi-Tech Industries PLC

ACADEMIC REFERENCES

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